

1 Sharing sunlight between trackers and crops: dynamic
2 light-sharing control for agrivoltaics under a
3 food-security floor on the world's staple croplands

4 Quang-Vinh Dang*

5 *British University Vietnam, Hung Yen, Vietnam*

6 **Abstract**

Single-axis trackers let agrivoltaic arrays shift the daily rotation from maximum electricity towards letting light through (light sharing). We quantify its value under a food-security floor of 90% of open-field yield. A differentiable simulator couples solar geometry, trackers, PV output, crop light, microclimate, soil water and the SIMPLE crop model. The crop response to shade is its least certain part, so we learned it from data: a hierarchical model fitted to 428 observations from 58 field studies predicts a new study of the four staple crop types only to about 0.22 in relative yield, no better than proportional loss, so between-study spread is large; gradient descent through the simulator turned the learned curves and four field trials into an ensemble of plausible responses. In a deployable protocol (density chosen from each site's history, held-out sites and years, 400 crop-site pairs), light-sharing rules allowed arrays three times as dense as static arrays at equal retention and raised the land-equivalent ratio (LER) from 1.07 to 1.18–1.24; a phenology rule needing no forecast was the best. A neural controller trained through the simulator added only 0.01–0.05 LER, and neither training on the ensemble nor a farm-measurable biomass signal changed this. The decision that mattered was the design: a density chosen under the mean response left 67–70% of sites at the floor when another plausible response held (oracle 83–87%), and a density meeting the floor in 75% of plausible responses raised this to 77–78% (+8 to +10 points, 95% intervals 5–14) for 0.04 LER, about half of the gap to the oracle. The simulator is too mild for maize and soybean, so gains for these crops are probably overestimated. Equipping 5% of staple cropland would give about 14 PWh yr^{-1} but needs about 10 TWp of modules.

7 **Keywords:** Agrivoltaics, Light sharing, Solar tracking, Food–energy nexus,

9 **1. Introduction**

10 Two of the defining challenges of this century compete for the same re-
11 source: land. The energy transition requires terawatts of new solar pho-
12 tovoltaic (PV) capacity, and ground-mounted solar farms need roughly one
13 to two hectares per megawatt. Food production must rise while crops are
14 exposed to more frequent heat waves and droughts. Warming alone is pro-
15 jected to reduce global yields of maize, wheat, rice and soybean by several
16 per cent per degree (Zhao et al., 2017), and a new generation of climate and
17 crop models shows these impacts emerging earlier than previously expected
18 (Jägermeyr et al., 2021). Croplands are also where solar power potential is
19 greatest (Adeh et al., 2019). They are flat, sunny, accessible and close to
20 rural electricity demand, so converting farmland into solar farms is tempting
21 and controversial at the same time.

22 Agrivoltaic (AV) systems combine crops and PV modules on the same
23 land and aim to resolve this conflict (Dupraz et al., 2011; Dinesh and Pearce,
24 2016; Weselek et al., 2019). By sharing sunlight between modules and plants,
25 a hectare can deliver more food plus energy than the two uses side by side,
26 with land-equivalent ratios (LER) well above one in field trials and mod-
27 elling studies (Dupraz et al., 2011; Amaducci et al., 2018; Valle et al., 2017;
28 Trommsdorff et al., 2021). The partial shade also changes the crop micro-
29 climate. It lowers canopy temperature and evaporative demand, conserves
30 soil moisture and improves water-use efficiency, especially in drylands (Mar-
31 rou et al., 2013b; Hassanpour Adeh et al., 2018; Barron-Gafford et al., 2019;
32 Elamri et al., 2018; Weselek et al., 2021). Agrivoltaics is therefore a candi-
33 date adaptation measure against heat and drought stress, in addition to a
34 mitigation measure through renewable electricity. The same shade, however,
35 also reduces the light available for photosynthesis. Regulators increasingly
36 require AV systems to retain most of the reference crop yield. The German
37 pre-standard DIN SPEC 91434, for example, requires at least 66% of the
38 reference yield (DIN, 2021), and other schemes and many farmers set stricter
39 floors. How much electricity can be produced without sacrificing food secu-
40 rity depends on the crop, the local climate, the year and the growth stage.

41 Most AV designs fix the geometry of the array, i.e. the ground coverage
42 ratio (GCR, module area per ground area) and the tilt or tracking rule, from

43 generic rules of thumb. Single-axis trackers add a degree of freedom: the
44 modules can be rotated away from the sun to let light through when the crop
45 needs it, as demonstrated for lettuce by Valle et al. (2017), or towards the
46 sun to maximise electricity. Choosing the rotation every day is a sequential
47 decision problem. The value of light depends on the crop’s phenological stage,
48 soil water and upcoming weather, and the effect of a decision appears in the
49 harvest months later. Recent work in this journal has used evolutionary
50 multi-objective optimisation to lay out PV arrays on farmland (Zhang et al.,
51 2026). Machine learning for crop management has mainly used reinforcement
52 learning with black-box simulators (Gautron et al., 2022), which is sample-
53 inefficient and hard to scale to global, multi-decadal assessments.

54 Two questions follow. First, how much of the electricity that an agri-
55 voltaic system can produce under a food-security floor comes from the array
56 density, and how much from dynamic light-sharing control of the trackers?
57 Second, how robust is that answer to the weakest part of any such study, the
58 crop response to shade, and does a controller learned from a simulator add
59 anything to a well-designed hand-written rule? Answering them requires
60 a simulator that couples the physics of light, PV, microclimate and crop
61 growth, a protocol in which all controllers face the same design constraints,
62 a check of the shade response against field data, and enough sites and seasons
63 to separate real effects from noise.

64 This paper makes three contributions. The first is a quantification of
65 light-sharing control under a yield floor. Light sharing is parameterised by
66 one daily level between full backtracking and the rotation that minimises the
67 shadow on the crop, and we compare static arrays, a seasonal schedule, a rule
68 that follows crop phenology, a rule that reacts to heat and drought stress,
69 and a neural controller (SALS) trained through the simulator. The second
70 is a deployable evaluation protocol. The array density of every site is chosen
71 from its historical seasons only, with the same rule for every method, and
72 performance is measured on held-out sites and years, over three training seeds
73 and three spatial splits. The third is a differentiable agrivoltaic simulator
74 that supports both, coupling hourly solar geometry, single-axis trackers with
75 backtracking, row-to-row shading and PV electricity, the direct and diffuse
76 light reaching the crop, canopy temperature and reference evapotranspiration
77 under the modules, and the SIMPLE crop model (Zhao et al., 2019) extended
78 with a flowering-stage shade penalty. All components are differentiable tensor
79 operations (Karniadakis et al., 2021). The PV part is benchmarked against
80 PVGIS. The crop response to shade is learned from the open data of a meta-

analysis (Laub et al., 2022), checked against field trials on rice, wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-Fuentes et al., 2023), and propagated through the simulator as an ensemble.

We apply the protocol to 400 crop–site pairs that represent where maize, wheat, rice, soybean and potato are grown worldwide, sampled in proportion to their harvested area (Yu et al., 2020) with local crop calendars (Jägermeyr et al., 2021), and driven by NASA POWER daily weather for 2001–2019 (Sparks, 2018) under baseline and +1.5, +2 and +3 °C climates. The main findings are as follows.

1. Light sharing is the main lever. At the same mean yield retention of 90%, arrays with light-sharing control carry about three times the module area of static arrays and produce 1.8–2.4 times as much electricity (Section 3.1).
2. The best hand-written rule follows crop phenology and needs no weather forecast. Rules that react to heat or drought stress are not better and depend on forecast quality (Section 3.3).
3. The crop response to shade can be learned from data. A hierarchical model fitted to 428 observations from 58 field studies predicts a held-out study of the four staple crop types to about 0.22 in relative yield, no better than proportional loss (it is better only for crop types the simulator does not represent), so between-study spread is large. The simulator is close to these curves for cereals and potato but too mild for maize and soybean, so the gains for those two crops are probably overestimated (Section 3.7).
4. A learned controller adds only a small gain over the best rule, 0.01–0.05 LER. Training it on an ensemble of shade responses, or giving it a farm-measurable biomass signal, does not change this (Section 3.8).
5. The bottleneck is the uncertainty of the crop response that enters the design. A density chosen so that the floor holds in 75% of the plausible responses raises the share of compliant sites from 67–70% (design under the mean response) to 77–78% (oracle: 83–87%) for 0.04 LER, pooled over three spatial splits (Section 3.8).
6. At the yield floor, agrivoltaics on 5% of the world’s staple cropland could supply about 14 PWh yr^{−1}, but only with about 10 TWp of modules, several times the installed PV capacity of the world (Section 3.10).

116 2. Materials and methods

117 2.1. Global cropland sites, weather and crop calendars

118 We studied five staple crops that together cover about 670 Mha of har-
119 vested area and 580 Mha of physical cropland: wheat, rice, maize, soybean
120 and potato. Harvested areas (rainfed and irrigated) were taken from the
121 SPAM 2010 gridded production maps (Yu et al., 2020) and aggregated to 1°
122 cells. For each crop, 80 cells with at least 5000 ha of the crop were drawn
123 without replacement with probability proportional to harvested area. An un-
124 weighted average over the sample therefore approximates an area-weighted
125 average over global production. The 400 crop-site pairs lie in 381 distinct
126 cells in 50 countries (Fig. 2). Sowing and maturity dates came from the
127 GGCMI Phase 3 crop calendar at 0.5° (Jägermeyr et al., 2021). For wheat
128 we used the winter-wheat calendar unless the coldest-month mean temper-
129 ature was below -8°C , in which case the spring-wheat calendar was used.
130 Daily weather for 2001–2020 was retrieved from NASA POWER (Sparks,
131 2018): global and diffuse shortwave irradiance, maximum, minimum and
132 dew-point temperature, precipitation and wind speed at 2 m. Each site con-
133 tributes 19 seasons (sowing years 2001–2019), each simulated as the 365 days
134 after sowing, which gives 7600 site-seasons.

135 Climate change was represented by the global warming levels $+1.5$, $+2$
136 and $+3^\circ\text{C}$ relative to 2001–2020. Daily maximum, minimum and dew-point
137 temperatures were shifted uniformly, which keeps relative humidity approx-
138 imately constant and raises the vapour-pressure deficit. The atmospheric
139 CO_2 concentration was set to 390, 450, 500 and 600 ppm for the baseline and
140 the three warming levels. Precipitation and irradiance were left unchanged.
141 This delta approach isolates the thermal signal of warming; its limitations
142 are discussed in Section 4.5.

143 2.2. A differentiable agrivoltaic simulator

144 The simulator advances all site-seasons in parallel with a daily time step
145 for the crop and an hourly time step for radiation and PV (Fig. 1). All equa-
146 tions are implemented as PyTorch tensor operations (Paszke et al., 2019), so
147 that yields and electricity are differentiable with respect to the design and
148 control variables.

149 *Hourly irradiance.* Daily global and diffuse irradiance were disaggregated
150 into hourly values with the Collares-Pereira and Rabl (1979) and Liu and

151 Jordan (1960) ratios and renormalised to the daily totals. Beam irradiance
 152 on the horizontal is the difference between the two. Solar position follows
 153 the standard declination and hour-angle equations.

154 *Array geometry and PV electricity.* We consider elevated horizontal single-
 155 axis trackers with north–south axes. The axis height is 1.25 module widths,
 156 e.g. 2.75 m for the common 2.2-m single-module-high (1P) tracker. The ro-
 157 tation ϕ is limited to $\pm 60^\circ$. The ground coverage ratio g is the module width
 158 divided by the row pitch and is restricted to $g \leq 0.5$. For 2.2-m modules this
 159 keeps a pitch of at least 4.4 m, and a free strip of at least 2.2 m between the
 160 rows when the modules are horizontal, which is the stow position during field
 161 operations. Let ψ be the solar zenith angle projected onto the east–west ver-
 162 tical plane. The energy-optimal rotation with backtracking (Lorenzo et al.,
 163 2011) is $\phi_{\text{bt}} = \psi - \text{sgn}(\psi) \arccos(\cos \psi / g)$ when $\cos \psi < g$, and $\phi_{\text{bt}} = \psi$ oth-
 164 erwise. The *light-sharing* rotation ϕ_{sh} is the feasible angle closest to edge-on
 165 to the sun, which minimises the shadow on the crop. Given a light-sharing
 166 level $u \in [0, 1]$ chosen once per day, the rotation in every hour is

$$\phi = (1 - u) \phi_{\text{bt}} + u \phi_{\text{sh}}. \quad (1)$$

167 Plane-of-array irradiance combines three terms: beam irradiance on the mod-
 168 ule, reduced by the fraction of the module shaded by the neighbouring row,
 169 $\max\{0, 1 - \cos \psi / (g |\cos(\phi - \psi)|)\}$; isotropic sky diffuse; and ground-reflected
 170 light (albedo 0.2). The cell temperature follows the NOCT model (45 °C),
 171 with an hourly air temperature interpolated sinusoidally between the daily
 172 extremes. Electricity per unit land is plane-of-array irradiance $\times \eta \times g$, with
 173 $\eta = 0.21 [1 - 0.0035(T_{\text{cell}} - 25)] (1 - 0.14)$ for module efficiency, temperature
 174 losses and system losses. The installed capacity is 2.1 g MWp ha⁻¹. A con-
 175 ventional ground-mounted PV plant ($g = 0.4$, backtracking, no crop) is the
 176 energy reference E_{PV} .

177 *Validation of the PV model.* We compared the simulated specific yield of an
 178 isolated tracker row ($g = 0.05$, negligible row-to-row shading) with PVGIS 5.2
 179 (European Commission, Joint Research Centre, 2022), for a horizontal north–
 180 south single-axis tracker with 14% losses, at 30 randomly chosen sites (29
 181 returned data). Across sites, the simulated and PVGIS specific yields were
 182 correlated with $r = 0.971$. The simulator was on average 4.9% lower (mean
 183 absolute deviation 5.5%). This is consistent with its $\pm 60^\circ$ rotation limit and
 184 with the different irradiance databases (NASA POWER vs. SARAH2, ERA5
 185 and NSRDB). Electricity estimates are therefore, if anything, conservative.

186 *Light and microclimate at the crop.* Averaged over the row pitch, the fraction
 187 of ground reached by beam light is $1 - \min\{1, g |\cos(\phi - \psi)| / \cos \psi\}$. The
 188 fraction of sky diffuse reaching the ground is a two-dimensional isotropic
 189 sky-view factor. We pre-computed it for 25 values of g and 13 rotation
 190 angles by ray casting over 31 neighbouring rows, and interpolate it bilinearly
 191 (differentiably). The daily ratio s of the irradiance at the crop to that of the
 192 open field defines the crop’s shading. The canopy maximum temperature is
 193 lowered by $\kappa(1 - s)$.

194 κ is uncertain. Field studies under PV modules report small reductions
 195 of daytime crop or canopy temperature and little change in air temperature
 196 (Marrou et al., 2013b; Barron-Gafford et al., 2019), and heat-stress thresholds
 197 calibrated on air temperature do not transfer directly to canopy temperature
 198 (Siebert et al., 2014; Webber et al., 2018). We therefore use a moderate
 199 $\kappa = 1^\circ\text{C}$ as the main case and report $\kappa = 0$ (no cooling) and $\kappa = 2^\circ\text{C}$
 200 for all methods (Section 3.2). Nocturnal warming under the modules is not
 201 represented. The reference evapotranspiration under the modules, ET_0 , is re-
 202 computed with the FAO-56 Penman–Monteith equation (Allen et al., 1998)
 203 from the reduced radiation and canopy temperature. Soil water conserva-
 204 tion under the modules (Hassanpour Adeg et al., 2018; Elamri et al., 2018)
 205 therefore emerges from the model, but rain redistribution by the modules
 206 (Marrou et al., 2013a) is ignored.

207 *Crop growth.* Crop growth follows the SIMPLE model (Zhao et al., 2019)
 208 with the species parameters of its Table 1a (Table 1). Daily biomass growth
 209 is

$$\Delta B = R_{\text{crop}} \cdot \text{RUE} \cdot f_{\text{Solar}} \cdot f_{\text{CO}_2} \cdot f_{\text{Temp}} \cdot \min\{f_{\text{Heat}}, f_{\text{Water}}\}, \quad (2)$$

210 where $R_{\text{crop}} = s R$ is the radiation received by the crop. The terms are as
 211 follows.

- 212 • f_{Solar} is the canopy’s radiation interception. In SIMPLE it depends
 213 only on thermal time, with senescence accelerated by heat and drought
 214 stress. Shade therefore reduces the daily biomass increment but not
 215 leaf-area development.
- 216 • f_{Heat} decreases linearly between the heat threshold T_{heat} and the ex-
 217 treme temperature T_{ext} of the canopy maximum temperature.
- 218 • $f_{\text{Water}} = 1 - S_{\text{water}} \text{ARID}$, where the agricultural reference index for
 219 drought (Woli et al., 2012) is computed from a daily soil-water balance

220 with curve-number runoff, drainage and root water uptake. Irrigated
 221 fields have $f_{\text{Water}} = 1$.

222 SIMPLE has a fixed harvest index and therefore ignores the high sen-
 223 sitivity of grain number to shading around flowering (Fischer, 1985; An-
 224 drade et al., 1999). It also ignores the differences in shade tolerance be-
 225 tween crop types. In the meta-analysis of Laub et al. (2022), maize and
 226 grain legumes lost yield strongly even at low shade levels, whereas C3 ce-
 227 reals and tuber crops initially lost less than proportionally; shading trials
 228 also show marked effects of shade on maize development (Ramos-Fuentes
 229 et al., 2023). We added a stage-dependent penalty. During a critical window
 230 ($0.45 \leq \text{TT}/T_{\text{sum}} \leq 0.65$), the harvest index is multiplied by $1 - h\bar{d}$, where $\bar{d}$
 231 is the mean relative radiation deficit $1 - s$ in that window. The sensitivity h
 232 is highest for maize (0.6) and soybean (0.5), the crop types that Laub et al.
 233 (2022) found most shade-sensitive, lower for the C3 cereals (wheat 0.4; rice
 234 0.5, for which no specific curve was available) and lowest for potato (0.2).
 235 These values are assumptions that reproduce this ranking; they are not cal-
 236 ibrated estimates. Section 3.6 shows the resulting shade-response curves.
 237 Because RUE is linear, SIMPLE does not credit the higher light-use effi-
 238 ciency of shaded canopies under more diffuse light. The penalty, conversely,
 239 makes shading around flowering more damaging than proportional. The net
 240 effect is closer to the field evidence than either assumption alone.

241 The thermal-time requirement to maturity was calibrated for every site as
 242 the mean thermal time between the GGCM sowing and maturity dates under
 243 the 2001–2020 climate, and the canopy parameters were scaled accordingly.
 244 The cultivar is thus adapted to the present climate, and warming shortens the
 245 season. Growth stops smoothly once the thermal-time requirement is met.
 246 Yield is biomass multiplied by the (shade-adjusted) harvest index. Each site
 247 is represented by a rainfed and an irrigated sub-field in proportion to its
 248 SPAM irrigated fraction. Each sub-field carries its own array, so both yield
 249 and electricity are area-weighted over the two sub-fields.

250 2.3. SALS: simulation-trained adaptive light sharing

251 SALS learns a *control network* $u_t = \pi_{\omega}(\mathbf{x}_t; g)$ (Algorithm 1). It is a
 252 multilayer perceptron with two hidden layers of 64 units and a bounded-logit
 253 output, $u = \sigma(5 \tanh(z/5)) \in [0.007, 0.993]$, which keeps gradients non-zero
 254 at the extremes. It chooses the daily light-sharing level of Eq. (1). Its inputs
 255 are the information available on day t :

- the crop and soil state: relative thermal time TT/T_{sum} , radiation interception f_{Solar} , relative soil water and the previous day's ARID;
- a *forecast* of the day's maximum temperature (relative to T_{heat} and T_{opt}), irradiance and ET_0 ;
- static information: the irrigation status of the sub-field, the array's GCR g , latitude, day of season and the crop.

Forecasts are perturbed with random errors, independently every day: Gaussian with a standard deviation of 2°C for maximum temperature, and multiplicative with standard deviations of 20% for irradiance and 15% for ET_0 . The physics always uses the true weather. The same forecast errors are applied to the rule-based controllers. The controller is conditioned on g . During training, each season is simulated with a GCR close to the density that is sustainable at its training site. This is the largest value on a coarse grid $(0.05, 0.10, \dots, 0.50)$ at which seasonal sharing keeps the mean retention of the site's training seasons at or above ρ , plus a uniform perturbation in $[-0.05, +0.10]$ truncated to $[0.05, 0.50]$. The network therefore learns to control the densities that design selection will later pick. Outside the cropping season the trackers backtrack for maximum electricity.

For site-season i with open-field yield Y_i^{open} , let $r_i = Y_i/Y_i^{\text{open}}$ be the yield retention and $e_i = E_i/E_{\text{PV},i}$ the electricity relative to a conventional PV plant. Training minimises the penalised objective

$$\mathcal{L}(\omega) = \mathbb{E}_{i \sim \mathcal{B}} \left[-e_i + \mu_1 \max\{0, \rho - r_i\} + \mu_2 \max\{0, \rho - r_i\}^2 \right], \quad \rho = 0.9, \mu_1 = 5, \mu_2 = 30, \quad (3)$$

which combines an exact (hinge) and a quadratic penalty for the food-security floor ρ . Because r_i and e_i are outputs of the differentiable simulator, the exact gradient of Eq. (3) is obtained by back-propagation through the 365 daily steps of every season (back-propagation through time). This is the key difference from reinforcement learning with black-box simulators (Gautron et al., 2022), which estimates gradients from sampled returns.

Each minibatch contains 512 site-seasons, each simulated as a rainfed and an irrigated sub-field, drawn from one of the four climates in turn. Seasons with a failed open-field crop ($Y^{\text{open}} < 0.2 \text{ t ha}^{-1}$) are excluded from training and evaluation. The network is fine-tuned through the simulator with Adam (Kingma and Ba, 2015) (learning rate 10^{-3} , cosine decay, gradient-norm clipping at 1) for 300 steps. Growth-stage logistic functions are evaluated in

Algorithm 1 SALS training

Require: training site-seasons in four climates; floor ρ ; penalties μ_1, μ_2

- 1: initialise π_ω by behaviour cloning of the phenology rule
- 2: **for** step = $1, \dots, N$ **do**
- 3: sample a climate and a minibatch $\mathcal{B}$ of site-seasons (rainfed and irrigated sub-fields)
- 4: set g_i to the site’s sustainable density plus a perturbation; initialise crop and soil states
- 5: **for** day $t = 1, \dots, 365$ **do**
- 6: $u_{it} \leftarrow \pi_\omega(\mathbf{x}_{it}; g_i)$ from state and noisy forecast if the crop is in the field, else 0
- 7: hourly rotation (Eq. 1), PV electricity, crop light, canopy temperature, ET_0
- 8: advance the soil-water balance and the SIMPLE crop state (Eq. 2)
- 9: $r_i \leftarrow Y_i/Y_i^{\text{open}}, e_i \leftarrow E_i/E_{\text{PV},i}$ (area-weighted); loss $\mathcal{L}$ (Eq. 3)
- 10: back-propagate through the season; Adam update of ω

289 a numerically stable form, so that gradients remain finite even under extreme
290 heat.

291 *Initialisation.* Trained from random weights, the controller converged to
292 sharing light throughout the cropping window ($u > 0.99$ on all days), i.e. to
293 the seasonal-sharing rule, because the gradient of electricity with respect to u
294 is small and of the same sign for most days of a season. We therefore initialise
295 the network by behaviour cloning of the phenology rule of Section 2.4: binary
296 cross-entropy against the rule’s output (soft targets 0.95 and 0.05) on about
297 1.7 million day-states visited by the rule in the training seasons, with 1500
298 Adam steps at a learning rate of 3×10^{-3} . Fine-tuning then moves the
299 policy away from this initialisation wherever doing so improves the objective.
300 Consequently, the gain of SALS over the phenology rule measures what fine-
301 tuning through the simulator adds to that rule, not what can be learned
302 without it.

303 2.4. Deployable design selection and baselines

304 A real array has one fixed density, which must be chosen before it is built.
305 Every method, SALS included, therefore uses the same *history-based design*
306 *rule*. For a site, the GCR is the largest value on the grid $\{0.05, 0.075, \dots, 0.50\}$
307 for which the method’s mean yield retention over the site’s 2001–2014 sea-

sons is at least $\rho = 0.9$. If no value qualifies, no array is built and the site remains open field. A stricter *chance-constrained* variant requires $r \geq 0.9$ in at least 80% of the historical seasons. The historical seasons of the test sites are never used for training, and the weather of the evaluation years is never used for design. We compare four controllers:

- *AV static*: backtracking trackers without light sharing ($u = 0$);
- *Seasonal sharing*: the light-sharing rotation ($u = 1$) throughout the cropping season;
- *Stress rule*: an expert rule that shares light ($u = 1$) during the main growth phase ($0.1 < TT/T_{\text{sum}} < 0.9$) except on days with a heat-stress forecast ($T_{\text{max}} \geq T_{\text{heat}}$) or drought ($\text{ARID} > 0.5$), when the modules track the sun and shade the crop;
- *SALS*: the learned controller.

The open field and a conventional PV plant without crops ($\text{LER} = 1$ each) are the references. Because the design rule is identical for all controllers, differences between methods reflect the quality of control and the array density that each controller can sustain under the food-security floor. We also compare the controllers at identical GCR over the whole grid (matched-density frontiers).

2.5. A shade-response model learned from field data

The simulator needs a crop response to shade that is supported by observations. We learned one from the open data of the meta-analysis of Laub et al. (2022) (Zenodo 10.5281/zenodo.5716091, CC BY 4.0): 428 observations of yield relative to an unshaded control at relative shade reductions of 0–90% from 58 studies, with crop and crop type, experiment type (artificial shading or intercropping), climate zone, irradiation and shade type. Relative yield is modelled as $y(s) = \exp(-\beta s^\gamma)$ with Student- t errors ($\nu = 4$; maximum a posteriori estimation, so this is a hierarchical shrinkage regression with bootstrap intervals and not a full Bayesian model). We compared six models: proportional loss $y = 1 - s$; one curve for all crops; one curve per crop type without pooling; one curve per crop type with partial pooling (Gaussian shrinkage of $\log \beta$ and $\log \gamma$ towards a common value, scale 0.5); partial pooling with covariates on $\log \beta$ (standardised irradiation, subtropical zone, solar-panel shade and intercropping); and gradient-boosted trees

on the same inputs. Fits use L-BFGS. Models were compared by leave-one-study-out cross-validation, in which every study is held out once, which tests the prediction of a new study and not of a new observation in a known study. Intervals for the crop-type curves and covariate effects come from a cluster bootstrap over studies (150 resamples).

2.6. Shade-response uncertainty: gradient calibration, robust controllers and design

The crop’s response to shade is the least certain part of the simulator, so we treat it as an unknown with a distribution. Two parameters per crop enter the simulator: a radiation-use compensation c (growth is multiplied by $1 + c(1 - s)$, s being the share of open-field irradiance that reaches the crop) and a multiplier h on the flowering-window harvest-index penalty. Because the simulator is differentiable, both can be fitted by gradient descent through whole simulated seasons to published yield–shading observations. The observations are the relative yields at 20% and 40% shading of the data-driven curves of Section 2.5 for the five crops (median, with an error equal to the width of the 80% bootstrap interval divided by 2.56 and at least 0.04; rice takes the curve of C3 cereals, which contains no rice), the field trials on rice, wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-Fuentes et al., 2023) (each season of each trial as a separate observation, error 0.06–0.08) and the unaffected harvest index of shaded maize (Ramos-Fuentes et al., 2023). We call the resulting ensemble E2. An earlier ensemble, E1, was fitted in the same way to curves read by hand from the figures of Laub et al. (2022) and was used to train the robust and adaptive controllers below. Priors are $c \sim \mathcal{N}(0.4, 0.5^2)$ and $h \sim \mathcal{N}(1, 0.5^2)$, and the parameters are clamped to $[0, 2.5]$ (c) and $[0, 2]$ (h) after every step, which is why a response milder than the data cannot be lowered further. We fitted 12 ensemble members by randomised maximum a posteriori estimation: every member sees the observations perturbed by their errors and a prior draw perturbed by its width, and is optimised with Adam (40 steps, with the loss still falling slightly at the last step; 40 training-block seasons per crop and member; shading of the static arrays cached). The ensemble is a device for generating plausible responses and not a validated posterior: it has no coverage check, and for some parameters (the harvest-index multiplier of wheat and potato) its spread equals that of the prior. The design analyses use five fixed members (indices 0, 3, 5, 8 and 10 of 12), chosen in advance. The ensemble is used in three ways.

379 *Robust and adaptive controllers.* SALS-robust has the architecture and
 380 training of SALS (Section 2.3), but every training season is simulated with the
 381 response of a randomly drawn ensemble member that the controller does not
 382 observe, and the density jitter is widened to $[-0.05, +0.25]$ because feasible
 383 densities differ between members. SALS-adaptive additionally observes the
 384 realised biomass of the crop under the modules relative to an open reference
 385 plot, a quantity that can be measured on the farm by remote sensing or by
 386 a reference plot. A feedback rule that shares light whenever this ratio falls
 387 below a target serves as the hand-written counterpart; its target (1.0) was
 388 chosen on training-block seasons under the ensemble.

389 *Uncertainty-aware design.* The design rule of Section 2.4 chooses the
 390 largest GCR whose historical retention meets the floor. Under shade-response
 391 uncertainty we choose the largest GCR that meets the floor in at least 75%
 392 of five ensemble members, and compare it with a design under the ensemble
 393 mean and with an oracle design under the true response. Evaluation of this
 394 rule is leave-one-out: the truth is the held-out member and the design uses
 395 the other four.

396 *Stress tests.* Controllers and rules are also evaluated under four fixed
 397 responses, the proportional response, the response calibrated to the meta-
 398 analysis, the response fitted to the field trials and a harsh response (propor-
 399 tional growth and a 1.5 times larger harvest-index penalty). The proportional
 400 and harsh responses lie outside the range of the ensemble.

401 2.7. Evaluation protocol, statistics and upscaling

402 Sites were split spatially: $10^\circ \times 10^\circ$ blocks were randomly assigned to train-
 403 ing (70%) or test (30%), so that no test site has a training site in the same
 404 block. Training uses the 2001–2014 seasons of the training sites. Design
 405 selection uses the 2001–2014 seasons of the test sites. Evaluation uses the
 406 2015–2019 seasons of the test sites. The same site-seasons are analysed in all
 407 four climates.

408 We report the following metrics:

- 409 • the mean electricity ($\text{MWh ha}^{-1} \text{yr}^{-1}$) and the relative electricity e ;
- 410 • the mean yield retention r and the production-weighted retention $\sum Y / \sum Y^{\text{open}}$;
- 411 • season-level compliance (share of seasons with $r \geq \rho$) and site-level
 412 compliance (share of sites whose mean test-season retention is $\geq \rho$);

Table 1: SIMPLE crop parameters (Zhao et al., 2019, Table 1a) and the harvest-index shade sensitivity h added in this study. T_{sum} , I_{50A} and I_{50B} are rescaled per site to the local crop calendar (Section 2.2). Units: temperatures in $^{\circ}\text{C}$, thermal times in $^{\circ}\text{C d}$, RUE in gMJ^{-1} ; S_{CO_2} is the relative RUE increase per 100 ppm above 350 ppm.

Crop (cultivar)	h	T_{sum}	HI	I_{50A}	I_{50B}	T_{base}	T_{opt}	RUE	$I_{50 \text{ max } H}$	$I_{50 \text{ max } W}$	T_{heat}	T_{ext}	S_{CO_2}	S_{water}
Wheat (Batten)	0.4	2150	0.34	280	50	0	15	1.24	100	25	34	45	0.08	0.4
Rice (IR72)	0.5	2300	0.47	850	200	9	26	1.24	100	10	34	50	0.08	1.0
Maize (McCurdy 84aa)	0.6	2050	0.50	500	50	8	28	2.10	100	12	34	50	0.01	1.2
Soybean (Williams 82)	0.5	2350	0.40	600	200	6	27	0.86	120	20	36	50	0.07	0.9
Potato (Sebago)	0.2	2400	0.85	500	350	4	22	1.30	50	30	34	45	0.10	0.4

- the lower tail (10th percentile of r , share of seasons with $r < 0.8$);
- the land-equivalent ratio $\text{LER} = r + e$ (Dupraz et al., 2011).

Statistical comparisons use site means as the unit of analysis. We report paired Wilcoxon signed-rank tests and cluster (site) bootstrap 95% confidence intervals of the mean difference. SALS was trained with three random seeds on the main split, and with one seed on each of two further spatial splits.

For the global estimate, every site is evaluated in the split in which it is held out. The mean electricity and yield per hectare of each crop are scaled to the *physical* area of that crop in SPAM 2010, which avoids double counting multiple-cropped land (Waha et al., 2020). We report the installed capacity (TWp), the capacity factor and a supply curve.

3. Results

3.1. Electricity and food on held-out sites and years

Table 2 compares the four controllers on the 92 held-out sites of the main split (seasons sown in 2015–2019), after each site’s density has been chosen from its 2001–2014 seasons with the same rule for all controllers. By construction all methods keep the mean yield retention close to 90% (0.904–0.917), so differences in electricity are differences in the density that each controller can sustain under this floor.

Static arrays are the weakest. Without light sharing, the floor limits the ground coverage to 6% on average, and arrays produce $217 \text{ MWh ha}^{-1} \text{ yr}^{-1}$, 16% of a conventional PV plant per hectare (LER 1.07). Rotating the modules to let light through during the cropping season (seasonal sharing) allows a coverage of 17% and $397 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ (LER 1.19). A rule

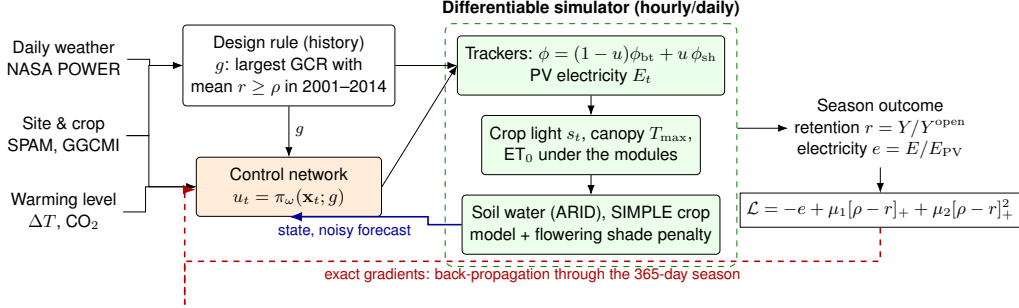


Figure 1: Simulator and controller. The array density g of a site is fixed from its historical seasons. A control network maps the daily crop and soil state, the density g and a noisy forecast to the light-sharing level u_t . The differentiable simulator converts these decisions into PV electricity, light and microclimate at the crop, soil water and crop growth. The season’s electricity and yield retention enter a penalised objective whose exact gradient is back-propagated through the 365-day season.

that shares light only while the crop is growing (phenology rule) reaches 465 MWh ha⁻¹ yr⁻¹ (LER 1.24). An expert rule that additionally shades the crop on forecast heat or drought days (stress rule) is worse than the phenology rule (LER 1.18): its density is limited to 13% because it reduces light at times when the floor is most binding.

The learned controller is best by a modest, consistent margin. SALS sustains a density of 17% (similar to the phenology rule), produces 510 MWh ha⁻¹ yr⁻¹ and reaches an LER of 1.270 at the same retention. Compared with the phenology rule this is +0.038 in LER (95% CI over sites 0.025–0.053; +10% electricity) and it is higher at 84% of the 92 sites (Table A.12). Compared with static arrays it is +0.205 (0.176–0.236), with 2.35 times the electricity at 2.9 times the density. The frontier in Fig. 3 shows the same picture at matched densities: the learned policy lies on or slightly above the phenology rule over the whole density range, both lie well above seasonal sharing and the stress rule, and static arrays are far below.

What the floor does and does not guarantee. Enforcing the floor on the historical mean does not guarantee it in every season. In the test seasons, 70% of SALS seasons have $r \geq 0.9$ (phenology rule 69%, stress rule 76%, static arrays 93%), and 2.7% have $r < 0.8$ (1.3%, 0.9% and 0.9%). The lower tail of SALS is therefore slightly heavier than that of the rules: its 10th percentile of r is 0.883, against 0.887, 0.886 and 0.903. Requiring $r \geq 0.9$ in at least 80% of a site’s historical seasons (chance-constrained

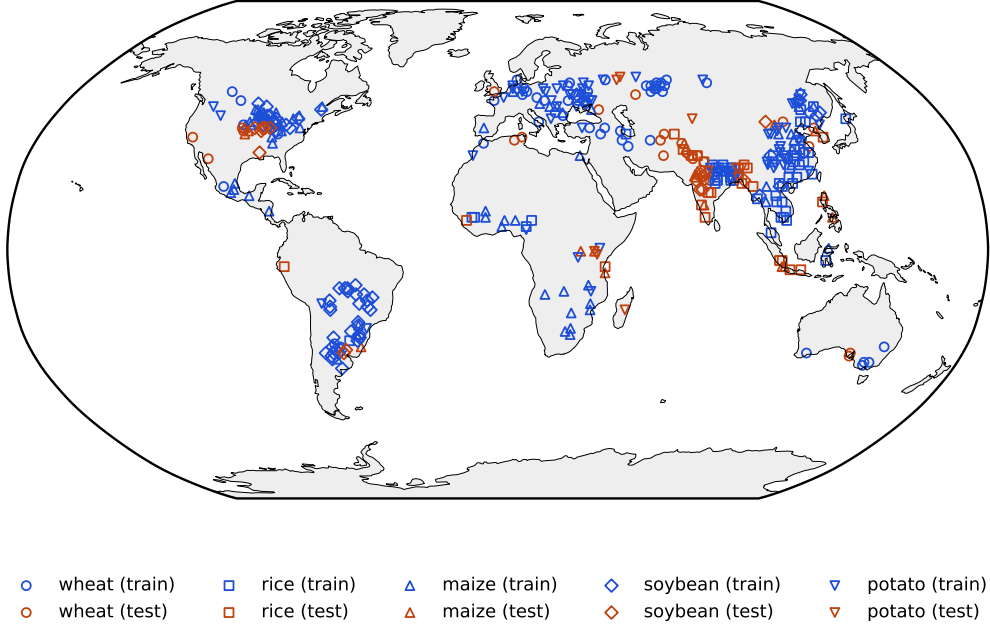


Figure 2: The 400 crop–site pairs, sampled in proportion to the harvested area of each crop (SPAM 2010). Blue: training sites; orange: geographically held-out test sites of the main split.

design) raises the season-level compliance to 90–94% for the light-sharing controllers. It costs about 0.065 in LER for SALS (1.205), which remains the best controller (phenology rule 1.192; Section 3.2).

3.2. Robustness

Seeds and splits. Three training seeds of SALS give LERs of 1.270, 1.268 and 1.269 under the baseline climate and 1.297, 1.298 and 1.297 at +3 °C. On two further spatial splits, in which different blocks of sites are held out, SALS exceeds the phenology rule by 0.047 and 0.041 in LER under the baseline climate and by 0.052 and 0.039 at +3 °C (main split: 0.034 and 0.032). In every split SALS ranks first, the phenology rule second and static arrays last.

Canopy cooling. The canopy-cooling coefficient κ is the most uncertain microclimate parameter. We repeated the whole protocol, including the historical design step, with $\kappa = 0, 1$ and 2 °C. Electricity changes with κ because cooler canopies tolerate denser arrays (SALS: 475, 510 and

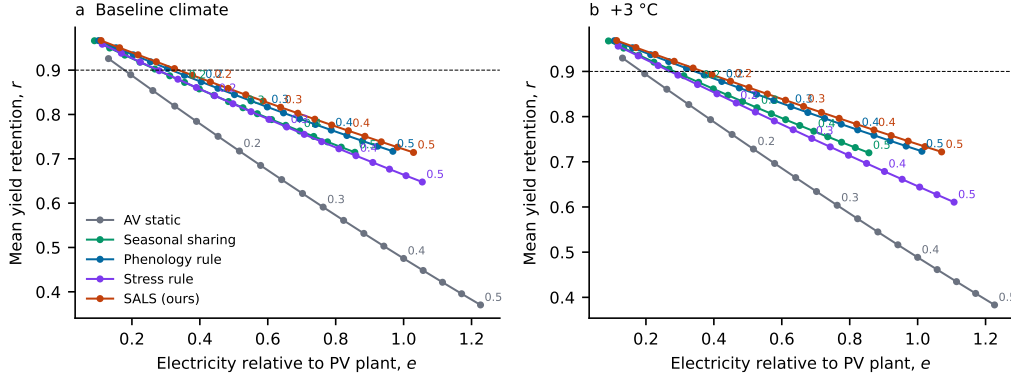


Figure 3: Electricity-retention frontiers at identical array densities (labels: GCR) for the held-out sites. Each point is the mean over test seasons at a fixed GCR; the dashed line is the 90% floor.

549 MWh ha⁻¹ yr⁻¹ under the baseline climate), but the retention is held at
 0.905 by the design rule and the ranking is unchanged. SALS exceeds the
 phenology rule by 0.029, 0.034 and 0.042 in LER under the baseline climate
 (0.022, 0.032 and 0.051 at +3 °C).

Which ingredients matter. Two ablations test the design of SALS
 at the same training budget. Masking the stress inputs (soil water, ARID,
 and the maximum temperature, irradiance and ET₀ forecasts) leaves the
 LER unchanged (1.269 vs 1.270 under the baseline climate, 1.297 vs 1.297
 at +3 °C). Training only on the baseline climate gives 1.268 and 1.291. The
 improvement over the rules therefore does not depend on stress-aware inputs
 or on training in warmed climates.

3.3. Floor conditioning, chance-constrained training and forecast errors

One network for all floors. Regulations and farmers set different
 floors. The network of the main experiments was trained at $\rho = 0.9$ only.
 We evaluated it at floors of 0.80, 0.85, 0.90 and 0.95, re-running the historical
 design step for each floor (Table 3). Its LER exceeds that of the phenology
 rule by 0.037, 0.036, 0.034 and 0.033 under the baseline climate (0.040, 0.037,
 0.032 and 0.033 at +3 °C), so the benefit carries over to other floors without
 retraining. A network that received the floor as an additional input and was
 trained over all four floors did not improve on it (differences in LER ≤ 0.001).
 Because the design step already adapts the density to the floor, conditioning
 the controller on the floor is not needed.

497 **Chance-constrained training.** We also trained a network whose ob-
 498 jective penalises a batch share of seasons below the floor that exceeds 20%.
 499 It did not improve the tail. With the mean-based design step its LER was
 500 1.263 (baseline) and 1.290 (+3 °C) against 1.270 and 1.297, and with chance-
 501 constrained design it reached season-level compliance of 92.6% against 92.0%
 502 and an LER of 1.198 against 1.205. The season-level guarantee is obtained
 503 by the design step (Section 3.1), not by the training objective.

504 **Forecast errors.** We raised the forecast errors seen by the controllers
 505 to 0, 1, 2 and 3 times the errors used in training. The LER of SALS and
 506 of the phenology rule was identical at all four error levels to three decimals
 507 (densities 0.2 and 0.3: SALS 1.307 and 1.457, phenology rule 1.278 and
 508 1.416). Neither controller uses the weather forecast: both rely on the crop’s
 509 thermal time, which is observable from on-farm temperature records. The
 510 stress rule, which depends on the forecast of maximum temperature, degrades
 511 with forecast error: at density 0.3 its retention falls from 0.778 to 0.751 and
 512 its LER from 1.432 to 1.418 as the errors triple.

513 3.4. *What the learned policy does*

514 Averaged over the rainfed test seasons of the main split at a density of 0.3,
 515 the light-sharing level of SALS is a smooth function of relative thermal time.
 516 It is close to zero in the first 10% of the thermal time required for maturity,
 517 rises through 0.55 at 10–20%, stays at 0.94–0.99 between 30% and 70%, and
 518 falls through 0.58 at 80–90% to about zero at maturity. The plateau covers
 519 the window in which our crop model penalises shading through the harvest
 520 index. The ramps at both ends let the modules produce electricity when
 521 additional light is worth least to the crop. Fig. 4 shows two example seasons.
 522 The phenology rule, which is also the initialisation of the network, shares
 523 light during the whole growing period. Fine-tuning through the simulator
 524 found that sharing can be reduced at the edges of the season: SALS gains its
 525 advantage by yielding light when it matters and collecting electricity when
 526 it matters less.

527 The policy also shares less light on days with a forecast heat stress: the
 528 mean level is 0.35 on such days against 0.60 (baseline) and 0.51 (+3 °C)
 529 on other days, and 58–63% of sites show a lower level on heat days. This
 530 cannot be read as a stress response, however. Heat days occur late in the
 531 season, when the phenological ramp is already falling, and a policy without
 532 any stress input shows a similar contrast (0.37 vs 0.57 in a separate check).

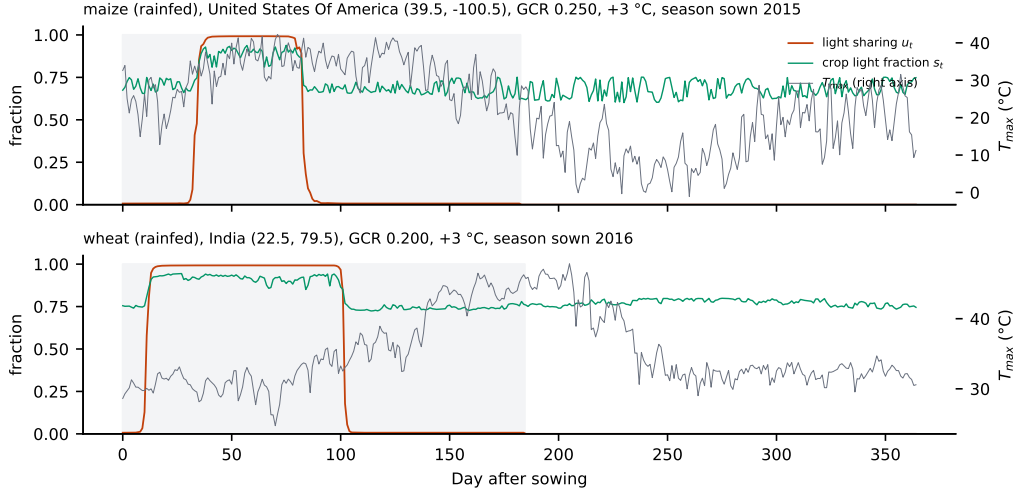


Figure 4: Learned daily control for two test seasons under $+3^{\circ}\text{C}$, at the density chosen for the site, simulated for the dominant sub-field of the site. Shading marks the cropping window. Orange: light-sharing level u_t ; green: fraction of open-field irradiance reaching the crop; grey: daily maximum temperature (right axis).

533 The ablation above confirms that stress information does not improve the
 534 result.

535 3.5. Crops and regions

536 The benefit of light sharing differs between crops (baseline climate, SALS).
 537 Soybean and maize tolerate the densest arrays (GCR 0.24 and 0.19; 691 and
 538 $577 \text{ MWh ha}^{-1} \text{ yr}^{-1}$; LER 1.41 and 1.30), potato follows (0.19; 542; 1.31),
 539 and rice and wheat are the most restrictive (0.13 and 0.15; 395 and 428; 1.20
 540 for both). The stress rule has a lower LER for every crop (1.26 for soybean,
 541 1.14–1.21 for the others).

542 The floor is not met evenly. Production-weighted retention is 0.888 for
 543 soybean and 0.898 for maize, below 0.9, because high-yielding seasons lose
 544 more than low-yielding ones. Regionally, Latin America and Oceania (4 sites)
 545 combine the highest density (GCR 0.41) and electricity ($1090 \text{ MWh ha}^{-1} \text{ yr}^{-1}$)
 546 with a production-weighted retention of only 0.862. Africa and the Middle
 547 East (8 sites) reach 0.894, and only 63% of their sites keep a mean retention
 548 of at least 0.9. South Asia, the region with the most sites (39), reaches 0.902
 549 (67% of sites) at a density of 0.14. A regulatory floor enforced per site would
 550 therefore need a region- and crop-specific design step, not a global one.

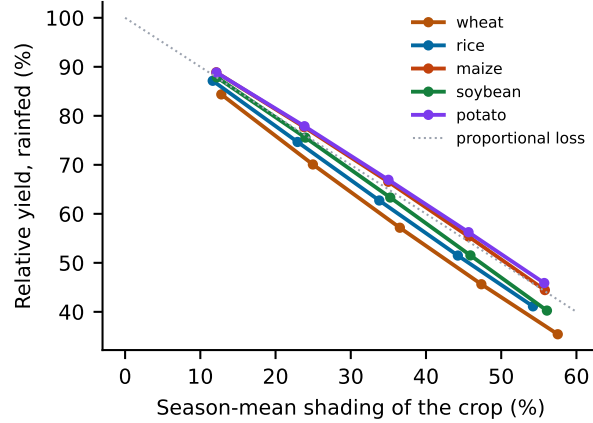


Figure 5: Simulated relative yield of rainfed crops under static arrays (GCR 0.1–0.5) against season-mean shading of the crop, baseline climate, held-out sites. The dotted line is proportional loss.

3.6. Model behaviour and validation

PV output. The simulated specific yield of an isolated tracker row agrees with PVGIS 5.2 (European Commission, Joint Research Centre, 2022) at 29 sites with $r = 0.971$, a mean bias of -4.9% and a mean absolute deviation of 5.5% (Section 2.2).

Shade response. Fig. 5 shows the simulated relative yield of rainfed crops under static arrays as a function of the season-mean shading of the crop. Yield declines close to proportionally with shading: 84–89% relative yield at 12% shading, 70–78% at 25%, 57–67% at 37% and 35–46% at 57%. Wheat loses most and potato and maize least. This response is steeper than the *less than proportional* losses that Laub et al. (2022) report for small shade levels of C3 cereals and tubers, and it does not reproduce their finding that maize and grain legumes are more shade sensitive than C3 cereals. In the simulator, shading also relieves heat and drought in rainfed maize and soybean, which offsets the higher harvest-index penalty we assigned to them. The absolute effect of shading on each crop is therefore uncertain (Section 4.5).

Comparison with field trials. We fitted one radiation-use compensation coefficient per crop to the yield–shading curves of Laub et al. (2022) using training-block sites only (Methods, calibration script in the repository). The fit changed wheat and rice only (coefficients 0.4 and 0.1) because the proportional model already lies within about 0.02 of the C3-cereal and tu-

ber curves, and it cannot lower the maize and legume responses below the proportional case (root-mean-square error 0.16 for maize and 0.08 for soybean). Against independent field trials (Table 4) the model underestimates retained yield: rice 0.79 against 0.84 (Gonocruz et al., 2021), wheat and potato 0.74 and 0.72 against 0.81–0.82 in the worse season of Weselek et al. (2021) (and above 1 in the better one), and maize 0.70–0.75 against 0.83–0.91 at 30–35% shading (Ramos-Fuentes et al., 2023), where harvest index was not affected by shade. The model is therefore conservative for food retention with respect to these trials, except for the stronger maize losses of the meta-analysis. Table 5 shows the consequence for the design results: with the calibrated response the allowed density and the LER of every method rise (phenology rule 1.28 against 1.23), and with a response fitted to the field trials the LER of the phenology rule is 1.62. The phenology rule is the best method in all three cases, and light sharing adds LER +0.17, +0.19 and +0.30 over static arrays (conservative, calibrated, field-fitted); only the order of the seasonal and stress rules changes. The absolute gains reported in this paper are thus lower bounds with respect to these trials and not upper bounds.

3.7. A shade-response model learned from field data

Cross-validation by study shows what the data can and cannot support (Tables 6 and 7). Over all 339 shaded observations the root-mean-square error of a held-out study falls from 0.43 for proportional loss $y = 1 - s$ to 0.37 for one curve for all crops and to 0.31 for crop-type curves, but this gain comes from crop types that the simulator does not represent (berries, forages, fruits and vegetables: 0.53 to 0.36). For the four crop types that the simulator uses, the learned curves are not better than proportional loss: the error is 0.222 against 0.220 pooled (difference +0.002, 95% interval -0.023 to $+0.030$), 0.187 against 0.214 for C3 cereals, and slightly worse for maize, grain legumes and tubers. Partial pooling is as good as separate curves (0.312 against 0.313) and the covariates add almost nothing (0.311). Gradient-boosted trees on the same inputs are worse (0.321) than the parametric model. The error of 0.22 for the four crop types is large: yields of a new study differ from the learned curve by about 20 percentage points, which is the between-study spread that a design under uncertainty has to cover. Crop type is the information that transfers between studies, and none of the covariate effects on $\log \beta$ is clearly different from zero (irradiation -0.10 , 90% interval -0.32 to 0.09 ; subtropical zone -0.37 , -0.73 to 0.07 ; intercropping 0.16 , -0.48 to 0.45).

609 The estimated effect of solar-panel shade is not interpretable, because all 46
610 panel observations are of leafy vegetables, mainly lettuce, and the effect is
611 confounded with crop type.

612 The learned curves show a clear ordering of crop types (Table 8). At
613 20% shading C3 cereals and tubers lose less than proportionally (relative
614 yield 0.93), whereas maize (0.73) and grain legumes (0.71) lose more; at 40%
615 shading the values are 0.72 and 0.71 against 0.42 and 0.40. Intervals are wide
616 for maize (0.27–0.59 at 40%) and for tubers, which rest on two studies. The
617 simulator’s responses are close to the data for cereals and potato but, under
618 both calibrated ensembles, too mild for maize and soybean at 40% shading
619 (0.75 and 0.48 for E2 against 0.42 and 0.40 in the data). The calibration
620 cannot lower the maize response further, because the simulated shading also
621 relieves heat and drought stress and the compensation and harvest-index pa-
622 rameters are bounded below at zero. The field trials of maize (Ramos-Fuentes
623 et al., 2023) are also milder than the meta-analysis. For maize and soybean
624 the gains reported in this paper are therefore probably overestimated.

625 3.8. Learning under shade-response uncertainty

626 **Calibration.** Gradient descent through the simulator fitted the five crops
627 to the data-driven curves and the field trials (ensemble E2, Table 9). The
628 data identify two features: the harvest index of maize is hardly affected by
629 shade (multiplier 0.16 ± 0.13 , $c \approx 0.14$), as observed by Ramos-Fuentes et al.
630 (2023), and soybean is more shade sensitive than the cereals (compensation
631 0.05, harvest-index multiplier 1.66), as in Laub et al. (2022). Wheat has the
632 largest compensation (0.91 ± 0.31) and the largest uncertainty.

633 **Learned controllers.** Controllers were trained on E1 and evaluated
634 under four fixed responses. Under every response the learned controller stays
635 better than the best hand-written rule, but only slightly (Table 10): the LER
636 gain of SALS over the phenology rule is +0.014, +0.015, +0.033 and +0.010
637 under the proportional, calibrated, field-fitted and harsh responses. Training
638 on the ensemble did not help: SALS-robust differs from SALS, which was
639 trained on the proportional response only, by at most 0.005 in LER, and
640 SALS-adaptive, which sees the measured biomass ratio, by at most 0.008.
641 The feedback rule is worse than the phenology rule by 0.025–0.064 in LER.
642 A plausible reason is that the biomass ratio does not reveal the harvest-index
643 loss from shading around flowering, so the information that the controller
644 could use to adapt is not in the measurement.

645 **Design under uncertainty.** The larger risk is in the design step (Ta-
646 ble 11). Pooled over the three spatial splits (93 held-out sites), a density
647 chosen under the mean response of E2 leaves 70% (phenology rule) and 67%
648 (SALS) of the sites at the floor when the true response is another plausible
649 one, against 87% and 83% for an oracle design. Choosing the density so
650 that the floor is met in at least 75% of the plausible responses raises site
651 compliance to 78% and 77% (change +7.9 (+4.9 to +11.4) and +9.6 (+6.0
652 to +13.5) percentage points, 95% cluster-bootstrap intervals over sites), at
653 a cost of 0.037 and 0.043 in LER. It recovers about half of the gap to the
654 oracle design, which remains +9.1 (+4.5 to +13.5) and +6.9 (+1.9 to +11.6)
655 points higher. On the first split alone, with the earlier ensemble E1, the same
656 design restored the oracle level of compliance (76% and 74%, against 58%
657 and 55% under the mean response) at a cost of 0.029 and 0.023 LER. For
658 the proportional and harsh responses, which lie outside the ensembles, the
659 design does not restore compliance (about 20–24% of sites on split 0), so the
660 guarantee holds only within the range of responses that the data support.

661 *3.9. Agrivoltaics under warming*

662 Under warming, the open-field production of the held-out sites (Fig. 6)
663 falls at +3 °C by 23% for maize and 18% for rice, is nearly unchanged for soy-
664 bean, and rises for wheat (+3%) and potato (+10%). Agrivoltaic production
665 with light-sharing control follows the open field at about 90% of its level, so
666 it inherits the same climate impacts (maize −31%, rice −27% relative to the
667 baseline open field at +3 °C). Agrivoltaics does not protect staple production
668 against warming in this model. The LER of light-sharing systems increases
669 slightly with warming (SALS 1.270 to 1.297; phenology rule 1.236 to 1.265),
670 because the floor allows slightly denser arrays in warmer conditions (GCR
671 0.173 to 0.177 and electricity 510 to 540 MWh ha^{−1} yr^{−1}). The margin of
672 SALS over the rules does not grow with warming (Table A.12).

673 *3.10. Global potential and what it would take*

674 Every site is evaluated when it is held out (219 of the 400 crop-site
675 pairs, in at least one split), and each crop’s mean electricity per hectare
676 is scaled to its physical area in SPAM 2010 (581 Mha for the five crops,
677 which avoids counting multiple-cropped land twice). Equipping 5% of this
678 area (29.1 Mha) with SALS arrays at the floor would produce 14.4 PWh yr^{−1}
679 under the baseline climate and 16.0 PWh yr^{−1} at +3 °C, with a production-
680 weighted retention of 0.903 and 0.900. The phenology rule would give 13.0

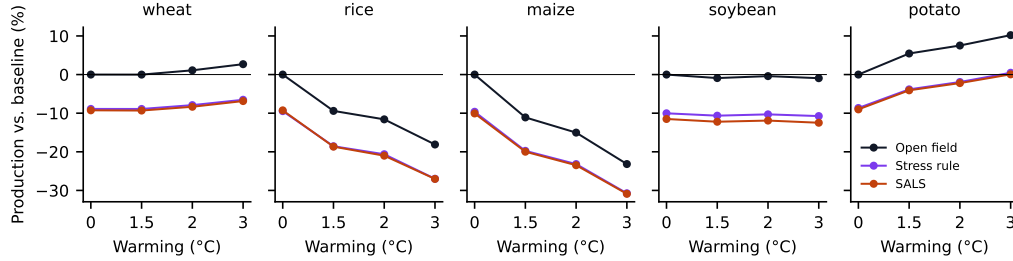


Figure 6: Production relative to the baseline open field (sum over held-out site-seasons) as a function of warming, for the open field, the stress rule and SALS.

681 and 14.5 PWh yr^{-1} , and static arrays 6.4 and 6.9 PWh yr^{-1} . For comparison,
682 world electricity generation was about 30 PWh in 2023 (Ember, 2024).

683 These numbers require large amounts of hardware. SALS at 5% cover-
684 age needs 10.6 TWp of modules (11.3 TWp at $+3^\circ\text{C}$), at a capacity factor of
685 15.5% (16.2%). World solar PV capacity at the end of 2024 was about 1.9 TW
686 (IRENA, 2025), so the requirement is about 5.7 times the installed PV fleet.
687 The electricity is also variable, peaks at midday and would be curtailed or
688 stored. The figures are technical potentials, not projections. Ordering sites
689 by electricity per hectare gives a supply curve (Fig. 7): the best 2% of the staple
690 cropland would supply 16 PWh yr^{-1} and the best 5% 40 PWh yr^{-1} with
691 SALS. The best tenth of the sites are mostly rainfed maize and soybean sites
692 in the Americas and in southern and eastern Africa (7% mean irrigated frac-
693 tion against 29% overall), which are also among the regions where the floor
694 is hardest to meet (Section 3.5). Siting by electricity alone would therefore
695 concentrate agrivoltaics where food-security risk is highest.

696 4. Discussion

697 4.1. What the results establish

698 Four findings stand on consistent evidence. First, sharing light with the
699 crop is the main lever on the electricity that tracker-based agrivoltaic arrays
700 can produce under a yield floor. With static trackers the floor limits arrays
701 to about 6% ground coverage and an LER of 1.07. Any light-sharing control
702 allows about three times the density, and the LER rises to 1.18–1.24 for hand-
703 written rules. Second, the best simple rule follows crop phenology. It shares
704 light across the middle of the season, around flowering and grain filling, and
705 produces electricity at the beginning and the end. It needs only the crop’s

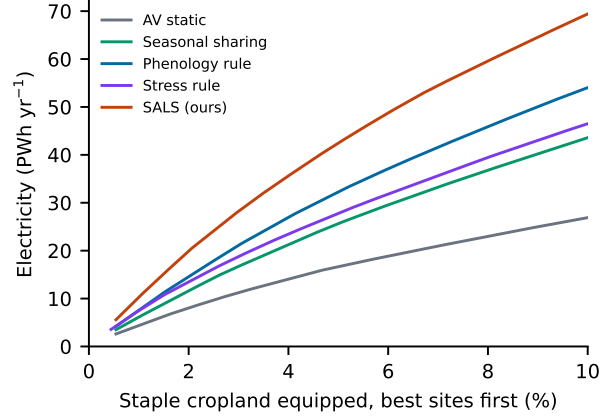


Figure 7: Supply curve under the baseline climate: cumulative electricity when sites are equipped in order of decreasing electricity per hectare, against the share of staple cropland (physical area) equipped. Each controller is evaluated at the density that keeps its mean retention at 0.9.

706 thermal time, which is observable on the farm, and it is insensitive to forecast
707 errors, whereas the rule that reacts to heat and drought stress is worse in
708 every climate and degrades with forecast error. Third, the qualitative result
709 does not depend on the weakest part of the model. The simulated shade
710 response is conservative against the field trials we could use (Section 3.6);
711 calibrating it raises the allowed density, and the phenology rule stays the
712 best method, with a gain over static arrays of 0.17, 0.19 and 0.30 LER
713 under the conservative, calibrated and field-fitted responses. However, the
714 response learned from the meta-analysis shows steeper losses for maize and
715 grain legumes than the simulator reproduces (Section 3.7), so the absolute
716 gains are lower bounds for cereals and potato and probably overestimates for
717 maize and soybean. Fourth, a controller learned through the differentiable
718 simulator is better than the best rule we could write, by 0.03–0.05 in LER and
719 about 10% in electricity at equal retention, reproducibly over training seeds
720 (range 0.002 in LER), three spatial splits, three canopy-cooling assumptions
721 and four climates, and at 84–86% of the held-out sites. This gain is small
722 compared with the gap between static and dynamic control.

723 4.2. What role does artificial intelligence play?

724 We tested three ways in which learning could matter, and the answers dif-
725 fer. The first is learning the crop’s response to shade from data. The learned

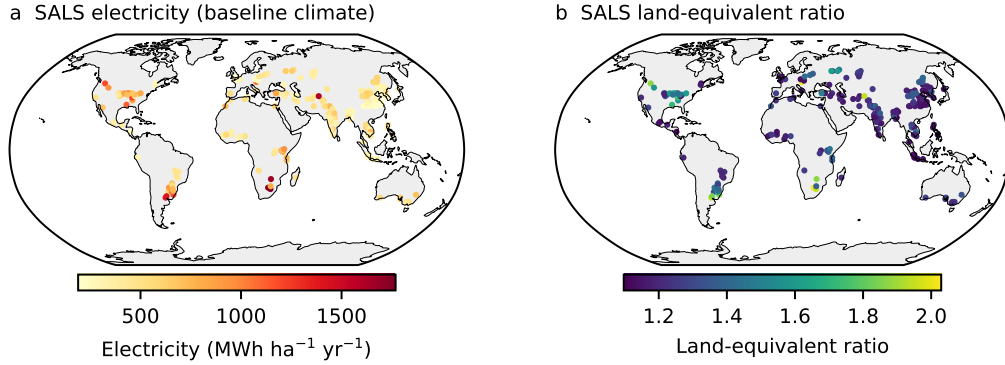


Figure 8: SALS electricity (a) and land-equivalent ratio (b) at the pooled held-out sites (baseline climate, mean of seasons 2015–2019).

726 model is useful mainly as a measurement of how little can be learned: a hier-
 727 archical model fitted to 428 observations from 58 studies predicts a held-out
 728 study of the four staple crop types with an error of 0.22 in relative yield, no
 729 better than proportional loss, and the gain over proportional loss (0.43 to
 730 0.31 over all types) comes from crop types that the simulator does not repre-
 731 sent. Crop type is the information that transfers, whereas irradiation, climate
 732 zone and shade type add nothing measurable and a non-parametric model
 733 (gradient-boosted trees) is worse with this number of studies (Section 3.7).
 734 Two things follow. The shade response of the staple crops is a distribution
 735 with a spread of about 20 percentage points between studies and not a curve,
 736 and the process-based simulator does not reproduce the steep losses of maize
 737 and legumes seen in the data. Learning the response was limited by the data
 738 and not by the model: two studies support the tuber curve, no rice study
 739 exists, and 46 panel observations are all leafy vegetables.

740 The second is calibration through the differentiable simulator. Gradient
 741 descent through whole seasons fitted the response of each crop to the learned
 742 curves and to the field trials, and it produced an ensemble of plausible re-
 743 sponses that reconciles them (Section 3.8). The ensemble drives the design
 744 decision: a density chosen under the mean response leaves 67–70% of sites at
 745 the floor if another plausible response holds, against 83–87% for an oracle,
 746 and a density chosen to meet the floor in 75% of plausible responses recov-
 747 ers about half of that gap (to 77–78%, pooled over three splits) at a cost of
 748 0.04 LER. The bottleneck of tracker-based agrivoltaics under a food-security

749 floor is therefore the quantified uncertainty of the crop response that enters
750 the design, and it is data on that response, not larger networks, that would
751 improve the result. The guarantee holds only within the range of responses
752 that the ensemble spans.

753 The third is control, and here learning contributes little. The learned
754 controller adds 0.01–0.03 LER over the best hand-written rule, which follows
755 crop phenology, under every response we tested, and a reader who needs a
756 simple, auditable controller can use that rule and obtain most of the electric-
757 ity gain over static arrays. Neither of the two extensions that we expected to
758 matter under uncertainty did: training on an ensemble of responses did not
759 change the policy, and neither did giving the controller a farm-measurable
760 signal, the biomass of the crop relative to a reference plot, nor a feedback
761 rule built on that signal. The learned policy also has to be initialised from
762 the phenology rule (Section 4.5). We therefore do not claim that learned
763 control is what makes agrivoltaics work: in this model, the schedule that a
764 learned policy converges to is a smooth function of crop phenology, which a
765 rule can already express.

766 *4.3. What the results do not establish*

767 The early hypothesis that stress-aware control, that is, shading the crop
768 on heat and drought days, would be a main source of benefit under warming
769 was not supported. The stress rule performs worse than the phenology rule
770 in every climate, and a learned controller without stress inputs performs as
771 well as one with them. A plausible reason is that in our crop model the yield
772 gain from shading on a hot or dry day is smaller than the loss of light on
773 that day, so stress information does not help. Whether this holds for crop
774 models with more detailed canopy-temperature and heat-stress responses is
775 an open question. Agrivoltaics does not protect staple production against
776 warming in our simulations: light-sharing systems produce about 90% of the
777 open-field yield in every climate and follow the same decline in maize and
778 rice (Fig. 6).

779 *4.4. Food security in practice*

780 Holding the historical mean retention at 90% does not hold it in each
781 season or at each site. 70% of SALS seasons meet the floor, the lower tail is
782 slightly heavier than for the rules, and production-weighted retention is below
783 0.9 for soybean (0.888) and maize (0.898) and in several regions. Chance-
784 constrained design raises season-level compliance to about 92% at a cost of

0.065 in LER, and the ranking of the controllers is preserved. Whichever formulation is chosen, a per-site regulatory floor would need a region- and crop-specific design step. The upscaled potential (Section 3.10) shows the trade-off in magnitude. The 14 PWh yr⁻¹ that 5% of the staple cropland could supply need about 10 TWp of modules, several times the present global fleet, and the best sites are rainfed maize and soybean areas where production-weighted retention is lowest.

4.5. Limitations

- *Simulation only.* All results come from a simulator. PV output was benchmarked against PVGIS (29 sites, mean bias -4.9%), but the crop response to shade was learned from one meta-analysis data set and compared with four field trials (Sections 3.6 and 3.7). The simulated response is conservative with respect to the trials (rice, wheat, potato, maize) but too mild for maize and soybean relative to the meta-analysis, where the 40%-shading relative yields are 0.42 and 0.40 against 0.75 and 0.48 in the simulator. The harvest-index sensitivities h are assumptions, and the sensitivity analysis uses rule controllers on a subsample of 22 sites.
- *Canopy cooling.* The effect of shade on canopy temperature was represented by one coefficient κ , applied to a heat-stress function that was calibrated on air temperature (Siebert et al., 2014; Webber et al., 2018). The ranking of the controllers is robust to κ between 0 and 2 °C, but the absolute electricity changes by about $\pm 8\%$. Nocturnal warming under modules and rain redistribution are not modelled.
- *Crop model.* SIMPLE has a fixed harvest index and linear radiation-use efficiency. More detailed models with grain-number formation and canopy energy balance may favour different schedules.
- *Climate scenarios.* Warming was applied as uniform temperature shifts with fixed precipitation, radiation and relative humidity, and with unchanged cultivars and sowing dates. It is not a bias-adjusted climate projection.
- *Initialisation and state information.* The network is initialised from the phenology rule, because training from random weights collapsed to full sharing. The gain of SALS is therefore the gain of fine-tuning that rule,

and we cannot say whether a better optimiser or a richer policy class would find larger gains. The controller uses the simulated state of the crop and soil (thermal time, canopy interception, soil water) without measurement error, and the forecast errors we tested apply to weather only.

- *Economics and deployment.* There is no cost, market or grid analysis. The value of midday electricity, curtailment, storage, structural cost and farmer acceptance (Trommsdorff et al., 2021) will determine the realisable share. The average row pitch at the chosen densities (about 13 m for 2.2-m modules) leaves room for machinery, but the densest designs (GCR up to 0.5) have a pitch of only 4.4 m.

5. Conclusions

Light sharing, the daily choice between maximum electricity and letting light through to the crop, is the main lever of tracker-based agrivoltaics under a food-security floor. In a differentiable simulator of trackers, PV output, crop light, microclimate and crop growth, evaluated fairly on 400 crop-site pairs with the array density of each site chosen from its own history, light-sharing rules allowed about three times the module density of static arrays at a mean yield retention of 90%. They raised the land-equivalent ratio from 1.07 to 1.18–1.24, and a rule that follows crop phenology and needs no weather forecast was the best of them. The simulated shade response is conservative against the field trials we could use, so these gains are lower bounds, and calibrating the response raised them without changing the ranking of methods. The crop’s response to shade, the least certain input, was learned from 428 field observations: a hierarchical model predicted a held-out study of the staple crop types no better than proportional loss (error 0.22), which shows the size of the between-study spread, and gradient descent through the simulator turned it into an ensemble of plausible responses. The simulator is too mild for maize and soybean, so the gains for those crops are probably overestimated. A learned controller added a further 0.01–0.05 LER by following a smooth phenology-dependent schedule that a rule can express; training it on an ensemble of shade responses or giving it a farm-measurable biomass signal added nothing. Where learning changed a decision was the design: choosing the array density so that the floor holds in 75% of the plausible responses raised the share of compliant sites from 67–70% to 77–78% (oracle

854 83–87%) at a cost of 0.04 LER, about half of the gap to an oracle design.
855 The yield floor is met on average but not in every season or region, and the
856 electricity potential of 5% of staple cropland (about 14 PWh yr⁻¹) requires
857 about 10 TWp of modules. Future work should validate the crop response
858 to shade under trackers with more field data, notably for maize and legumes
859 where the meta-analysis and the trials disagree, extend the crop model to
860 grain-number formation and canopy energy balance, add costs and grid con-
861 straints, and pursue controllers that guarantee the food-security floor per site
862 and per season.

863 **CRedit authorship contribution statement**

864 **Quang-Vinh Dang:** Conceptualization, Methodology, Software, Formal
865 analysis, Investigation, Data curation, Visualization, Writing – original draft,
866 Writing – review & editing.

867 **Declaration of competing interest**

868 The author has no known competing financial interests or personal re-
869 lationships that could have appeared to influence the work reported in this
870 paper.

871 **Data and code availability**

872 All data used are public: SPAM 2010 v2r0 harvested areas (Yu et al.,
873 2020), the GGCM Phase 3 crop calendar (Jägermeyr et al., 2021), NASA
874 POWER daily weather (Sparks, 2018) and the SIMPLE crop parameters
875 (Zhao et al., 2019). The code of the simulator, of SALS and of all experi-
876 ments, together with the site list, is available from the author’s public reposi-
877 tory (<https://github.com/vinhqdang/Artificial-Intelligence-in-Agriculture>).

878 **Funding**

879 This research did not receive any specific grant from funding agencies in
880 the public, commercial, or not-for-profit sectors.

881 **Appendix A. Statistical comparison**

882 **References**

- 883 Adeh, E.H., Good, S.P., Calaf, M., Higgins, C.W., 2019. Solar pv power
884 potential is greatest over croplands. *Scientific Reports* 9, 11442. doi:10.
885 1038/s41598-019-47803-3.
- 886 Allen, R.G., Pereira, L.S., Raes, D., Smith, M., 1998. Crop evapotranspira-
887 tion: Guidelines for computing crop water requirements. FAO Irrigation
888 and Drainage Paper 56. Food and Agriculture Organization of the United
889 Nations. Rome.
- 890 Amaducci, S., Yin, X., Colauzzi, M., 2018. Agrivoltaic systems to optimise
891 land use for electric energy production. *Applied Energy* 220, 545–561.
892 doi:10.1016/j.apenergy.2018.03.081.
- 893 Andrade, F.H., Vega, C., Uhart, S., Cirilo, A., Cantarero, M., Valentinuz, O.,
894 1999. Kernel number determination in maize. *Crop Science* 39, 453–459.
895 doi:10.2135/cropsci1999.0011183X0039000200026x.
- 896 Barron-Gafford, G.A., Pavao-Zuckerman, M.A., Minor, R.L., Sutter, L.F.,
897 Barnett-Moreno, I., Blackett, D.T., Thompson, M., Dimond, K., Gerlak,
898 A.K., Nabhan, G.P., Macknick, J.E., 2019. Agrivoltaics provide mutual
899 benefits across the food–energy–water nexus in drylands. *Nature Sustain-*
900 *ability* 2, 848–855. doi:10.1038/s41893-019-0364-5.
- 901 Collares-Pereira, M., Rabl, A., 1979. The average distribution of solar
902 radiation-correlations between diffuse and hemispherical and between daily
903 and hourly insolation values. *Solar Energy* 22, 155–164. doi:10.1016/
904 0038-092x(79)90100-2.
- 905 DIN, 2021. DIN SPEC 91434:2021-05: Agri-photovoltaic systems – require-
906 ments for primary agricultural use. Beuth Verlag, Berlin.
- 907 Dinesh, H., Pearce, J.M., 2016. The potential of agrivoltaic systems. *Renew-*
908 *able and Sustainable Energy Reviews* 54, 299–308. doi:10.1016/j.rser.
909 2015.10.024.
- 910 Dupraz, C., Marrou, H., Talbot, G., Dufour, L., Nogier, A., Ferard, Y., 2011.
911 Combining solar photovoltaic panels and food crops for optimising land

912 use: Towards new agrivoltaic schemes. *Renewable Energy* 36, 2725–2732.
913 doi:10.1016/j.renene.2011.03.005.

914 Elamri, Y., Cheviron, B., Lopez, J.M., Dejean, C., Belaud, G., 2018. Water
915 budget and crop modelling for agrivoltaic systems: Application to irrigated
916 lettuces. *Agricultural Water Management* 208, 440–453. doi:10.1016/j.
917 agwat.2018.07.001.

918 Ember, 2024. *Global Electricity Review 2024*. Technical Report. Ember.
919 London.

920 European Commission, Joint Research Centre, 2022. Photovoltaic geo-
921 graphical information system (PVGIS), version 5.2. https://re.jrc.ec.europa.eu/pvg_tools/.

923 Fischer, R.A., 1985. Number of kernels in wheat crops and the influence of
924 solar radiation and temperature. *The Journal of Agricultural Science* 105,
925 447–461. doi:10.1017/S0021859600056495.

926 Gautron, R., Maillard, O.A., Preux, P., Corbeels, M., Sabbadin, R., 2022.
927 Reinforcement learning for crop management support: Review, prospects
928 and challenges. *Computers and Electronics in Agriculture* 200, 107182.
929 doi:10.1016/j.compag.2022.107182.

930 Gonocruz, R.A., Nakamura, R., Yoshino, K., Homma, M., Doi, T., Yoshida,
931 Y., Tani, A., 2021. Analysis of the rice yield under an agrivoltaic
932 system: A case study in japan. *Environments* 8, 65. doi:10.3390/
933 environments8070065.

934 Hassanpour Adeg, E., Selker, J.S., Higgins, C.W., 2018. Remarkable agri-
935 voltaic influence on soil moisture, micrometeorology and water-use effi-
936 ciency. *PLOS ONE* 13, e0203256. doi:10.1371/journal.pone.0203256.

937 IRENA, 2025. *Renewable Capacity Statistics 2025*. Technical Report. Inter-
938 national Renewable Energy Agency. Abu Dhabi.

939 Jägermeyr, J., Müller, C., Ruane, A.C., Elliott, J., Balkovic, J., Castillo, O.,
940 Faye, B., Foster, I., Folberth, C., Franke, J.A., Fuchs, K., Guarin, J.R.,
941 Heinke, J., Hoogenboom, G., Iizumi, T., Jain, A.K., Kelly, D., Khabarov,
942 N., Lange, S., Lin, T.S., Liu, W., Mialyk, O., Minoli, S., Moyer, E.J.,
943 Okada, M., Phillips, M., Porter, C., Rabin, S.S., Scheer, C., Schneider,

- 944 J.M., Schyns, J.F., Skalsky, R., Smerald, A., Stella, T., Stephens, H.,
945 Webber, H., Zabel, F., Rosenzweig, C., 2021. Climate impacts on global
946 agriculture emerge earlier in new generation of climate and crop models.
947 *Nature Food* 2, 873–885. doi:10.1038/s43016-021-00400-y.
- 948 Karniadakis, G.E., Kevrekidis, I.G., Lu, L., Perdikaris, P., Wang, S., Yang,
949 L., 2021. Physics-informed machine learning. *Nature Reviews Physics* 3,
950 422–440. doi:10.1038/s42254-021-00314-5.
- 951 Kingma, D.P., Ba, J., 2015. Adam: A method for stochastic optimization,
952 in: *International Conference on Learning Representations*.
- 953 Laub, M., Pataczek, L., Feuerbacher, A., Zikeli, S., Högy, P., 2022. Contrast-
954 ing yield responses at varying levels of shade suggest different suitability
955 of crops for dual land-use systems: a meta-analysis. *Agronomy for Sus-
956 tainable Development* 42, 51. doi:10.1007/s13593-022-00783-7.
- 957 Liu, B.Y., Jordan, R.C., 1960. The interrelationship and characteristic dis-
958 tribution of direct, diffuse and total solar radiation. *Solar Energy* 4, 1–19.
959 doi:10.1016/0038-092x(60)90062-1.
- 960 Lorenzo, E., Narvarte, L., Muñoz, J., 2011. Tracking and back-tracking.
961 *Progress in Photovoltaics: Research and Applications* 19, 747–753. doi:10.
962 1002/pip.1085.
- 963 Marrou, H., Dufour, L., Wery, J., 2013a. How does a shelter of solar panels
964 influence water flows in a soil–crop system? *European Journal of Agron-
965 omy* 50, 38–51. doi:10.1016/j.eja.2013.05.004.
- 966 Marrou, H., Guilioni, L., Dufour, L., Dupraz, C., Wery, J., 2013b. Microcli-
967 mate under agrivoltaic systems: Is crop growth rate affected in the partial
968 shade of solar panels? *Agricultural and Forest Meteorology* 177, 117–132.
969 doi:10.1016/j.agrformet.2013.04.012.
- 970 Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen,
971 T., Lin, Z., Gimelshein, N., Antiga, L., et al., 2019. PyTorch: An impera-
972 tive style, high-performance deep learning library, in: *Advances in Neural
973 Information Processing Systems*, pp. 8024–8035.
- 974 Ramos-Fuentes, I.A., Elamri, Y., Cheviron, B., Dejean, C., Belaud, G.,
975 Fumey, D., 2023. Effects of shade and deficit irrigation on maize growth

976 and development in fixed and dynamic agrivoltaic systems. *Agricultural*
977 *Water Management* 280, 108187. doi:10.1016/j.agwat.2023.108187.

978 Siebert, S., Ewert, F., Eyshi Rezaei, E., Kage, H., Graß, R., 2014. Impact
979 of heat stress on crop yield—on the importance of considering canopy
980 temperature. *Environmental Research Letters* 9, 044012. doi:10.1088/
981 1748-9326/9/4/044012.

982 Sparks, A., 2018. nasapower: A nasa power global meteorology, surface solar
983 energy and climatology data client for r. *Journal of Open Source Software*
984 3, 1035. doi:10.21105/joss.01035.

985 Trommsdorff, M., Kang, J., Reise, C., Schindele, S., Bopp, G., Ehmann, A.,
986 Weselek, A., Högy, P., Obergfell, T., 2021. Combining food and energy
987 production: Design of an agrivoltaic system applied in arable and vegetable
988 farming in germany. *Renewable and Sustainable Energy Reviews* 140,
989 110694. doi:10.1016/j.rser.2020.110694.

990 Valle, B., Simonneau, T., Sourd, F., Pechier, P., Hamard, P., Frisson, T.,
991 Ryckewaert, M., Christophe, A., 2017. Increasing the total productivity of
992 a land by combining mobile photovoltaic panels and food crops. *Applied*
993 *Energy* 206, 1495–1507. doi:10.1016/j.apenergy.2017.09.113.

994 Waha, K., Dietrich, J.P., Portmann, F.T., Siebert, S., Thornton, P.K., Bon-
995 deau, A., Herrero, M., 2020. Multiple cropping systems of the world
996 and the potential for increasing cropping intensity. *Global Environmental*
997 *Change* 64, 102131. doi:10.1016/j.gloenvcha.2020.102131.

998 Webber, H., White, J.W., Kimball, B.A., Ewert, F., Asseng, S., Eyshi Rezaei,
999 E., Pinter, P.J., Hatfield, J.L., Reynolds, M.P., Ababaei, B., Bindi, M.,
1000 Doltra, J., Ferrise, R., Kage, H., Kassie, B.T., Kersebaum, K.C., Luig,
1001 A., Olesen, J.E., Semenov, M.A., Stratonovitch, P., Ratjen, A.M., LaM-
1002 orte, R.L., Leavitt, S.W., Hunsaker, D.J., Wall, G.W., Martre, P., 2018.
1003 Physical robustness of canopy temperature models for crop heat stress
1004 simulation across environments and production conditions. *Field Crops*
1005 *Research* 216, 75–88. doi:10.1016/j.fcr.2017.11.005.

1006 Weselek, A., Bauerle, A., Hartung, J., Zikeli, S., Lewandowski, I., Högy, P.,
1007 2021. Agrivoltaic system impacts on microclimate and yield of different

1008 crops within an organic crop rotation in a temperate climate. *Agronomy*
1009 for Sustainable Development 41, 59. doi:10.1007/s13593-021-00714-y.

1010 Weselek, A., Ehmann, A., Zikeli, S., Lewandowski, I., Schindele, S., Högy,
1011 P., 2019. Agrophotovoltaic systems: applications, challenges, and op-
1012 portunities. a review. *Agronomy for Sustainable Development* 39, 35.
1013 doi:10.1007/s13593-019-0581-3.

1014 Woli, P., Jones, J.W., Ingram, K.T., Fraisse, C.W., 2012. Agricultural
1015 reference index for drought (arid). *Agronomy Journal* 104, 287–300.
1016 doi:10.2134/agronj2011.0286.

1017 Yu, Q., You, L., Wood-Sichra, U., Ru, Y., Joglekar, A.K.B., Fritz, S., Xiong,
1018 W., Lu, M., Wu, W., Yang, P., 2020. A cultivated planet in 2010 – part 2:
1019 The global gridded agricultural-production maps. *Earth System Science*
1020 Data 12, 3545–3572. doi:10.5194/essd-12-3545-2020.

1021 Zhang, L., Geng, X., Ding, H., Cao, K., Wang, L., Chen, H., Deng, L., Wu,
1022 C., Xiao, M., Bao, E., 2026. Multi-objective optimization of photovoltaic
1023 array layouts on farmland via a hybrid algorithm framework for enhanced
1024 agricultural production and energy generation. *Artificial Intelligence in*
1025 *Agriculture* doi:10.1016/j.aiaa.2026.09.002.

1026 Zhao, C., Liu, B., Piao, S., Wang, X., Lobell, D.B., Huang, Y., Huang, M.,
1027 Yao, Y., Bassu, S., Ciais, P., Durand, J.L., Elliott, J., Ewert, F., Janssens,
1028 I.A., Li, T., Lin, E., Liu, Q., Martre, P., Müller, C., Peng, S., Peñuelas,
1029 J., Ruane, A.C., Wallach, D., Wang, T., Wu, D., Liu, Z., Zhu, Y., Zhu,
1030 Z., Asseng, S., 2017. Temperature increase reduces global yields of major
1031 crops in four independent estimates. *Proceedings of the National Academy*
1032 *of Sciences* 114, 9326–9331. doi:10.1073/pnas.1701762114.

1033 Zhao, C., Liu, B., Xiao, L., Hoogenboom, G., Boote, K.J., Kassie, B.T.,
1034 Pavan, W., Shelia, V., Kim, K.S., Hernandez-Ochoa, I.M., Wallach, D.,
1035 Porter, C.H., Stockle, C.O., Zhu, Y., Asseng, S., 2019. A simple crop
1036 model. *European Journal of Agronomy* 104, 97–106. doi:10.1016/j.eja.
1037 2019.01.009.

Table 2: Held-out performance (92 test sites, seasons 2015–2019, canopy cooling $\kappa = 1^\circ\text{C}$). The density (GCR) of each site is the largest value on the grid 0.05, 0.075, ..., 0.5 at which the method’s mean retention in the site’s 2001–2014 seasons is ≥ 0.9 . E : electricity ($\text{MWh ha}^{-1} \text{ yr}^{-1}$); e : electricity relative to a conventional PV plant; r : mean yield retention; r_{prod} : production-weighted retention; C_s and C_{site} : share of seasons with $r \geq 0.9$ and share of sites whose test-season mean $r \geq 0.9$ (%); $\text{LER} = r + e$.

Method	GCR	E	e	r	r_{prod}	C_s	C_{site}	LER
<i>Baseline</i>								
AV static	0.06	217	0.16	0.917	0.914	93	96	1.07
Seasonal sharing	0.17	397	0.28	0.904	0.902	69	72	1.19
Phenology rule	0.17	465	0.33	0.904	0.902	69	75	1.24
Stress rule	0.13	387	0.28	0.907	0.907	76	82	1.18
SALS (ours)	0.17	510	0.36	0.905	0.902	70	74	1.27
<i>+1.5 °C</i>								
AV static	0.06	221	0.16	0.916	0.914	93	92	1.08
Seasonal sharing	0.16	393	0.28	0.905	0.903	72	74	1.19
Phenology rule	0.17	478	0.34	0.905	0.902	68	75	1.25
Stress rule	0.12	377	0.27	0.908	0.908	73	77	1.18
SALS (ours)	0.17	518	0.37	0.906	0.903	68	75	1.28
<i>+2 °C</i>								
AV static	0.06	223	0.16	0.916	0.914	92	93	1.08
Seasonal sharing	0.17	394	0.28	0.905	0.904	72	76	1.19
Phenology rule	0.17	483	0.35	0.905	0.903	69	74	1.25
Stress rule	0.12	370	0.26	0.909	0.909	75	78	1.17
SALS (ours)	0.17	521	0.38	0.907	0.904	69	75	1.28
<i>+3 °C</i>								
AV static	0.06	231	0.17	0.915	0.914	88	89	1.08
Seasonal sharing	0.17	397	0.29	0.906	0.904	69	75	1.19
Phenology rule	0.18	499	0.36	0.905	0.902	67	72	1.26
Stress rule	0.11	366	0.26	0.908	0.908	74	77	1.17
SALS (ours)	0.18	540	0.39	0.907	0.904	68	75	1.30

Table 3: LER of the controllers at different yield-retention floors ρ (main split; the historical design step is repeated for each floor).

Method	Baseline climate				+3 °C			
	0.80	0.85	0.90	0.95	0.80	0.85	0.90	0.95
AV static	1.159	1.115	1.072	1.011	1.174	1.128	1.083	1.013
Stress rule	1.380	1.282	1.182	1.077	1.364	1.269	1.172	1.071
Phenology rule	1.467	1.356	1.236	1.108	1.513	1.395	1.265	1.122
SALS (trained at $\rho = 0.9$)	1.505	1.392	1.270	1.141	1.553	1.432	1.297	1.154
SALS conditioned on ρ	1.504	1.391	1.269	1.141	1.553	1.431	1.296	1.154

Table 4: Simulated relative yield at the shading of published field trials (static arrays, training-block sites). Observed values: rice, single trial; wheat and potato, the worse of the two seasons (the better season gave +2.7% and +11%); maize, mean of the three seasons of the dynamic system (DAV) and single values for the other two systems. The same trials enter the calibration of Section 2.6, so the comparison is not independent.

Trial	Shading	Observed	Conservative	Calibrated
Rice, Gonocruz et al. (2021)	0.25	0.84	0.77	0.79
Wheat, Weselek et al. (2021)	0.30	0.81	0.66	0.74
Potato, Weselek et al. (2021)	0.30	0.82	0.72	0.72
Maize, Ramos-Fuentes et al. (2023) (DAV)	0.35	0.83	0.70	0.70
Maize, Ramos-Fuentes et al. (2023) (half)	0.30	0.91	0.75	0.75
Maize, Ramos-Fuentes et al. (2023) (full)	0.50	0.70	0.54	0.54

Table 5: Sensitivity of the deployable light-sharing results to the crop shade response (rule controllers, baseline climate, floor $\rho = 0.9$, split 0, every fourth held-out site, 22 sites). Conservative: proportional response (main analysis); calibrated: radiation-use compensation fitted to Laub et al. (2022); field-fitted: fitted to the field trials, maize harvest index unaffected.

Shade response	Method	GCR	E/E_{PV}	Retention	LER
conservative	AV static	0.06	0.15	0.92	1.07
	Phenology rule	0.17	0.33	0.90	1.23
	Seasonal sharing	0.16	0.28	0.90	1.18
	Stress rule	0.11	0.24	0.91	1.15
calibrated	AV static	0.07	0.18	0.91	1.09
	Phenology rule	0.19	0.37	0.90	1.28
	Seasonal sharing	0.19	0.32	0.90	1.22
	Stress rule	0.13	0.29	0.91	1.20
field-fitted	AV static	0.16	0.41	0.91	1.32
	Phenology rule	0.37	0.70	0.92	1.62
	Seasonal sharing	0.37	0.61	0.92	1.53
	Stress rule	0.29	0.63	0.90	1.53

Table 6: Predicting relative yield under shade from 428 observations of 58 studies (Laub et al., 2022): leave-one-study-out cross-validation (every study held out once; observations with $s > 0$). Covariates: irradiation, climate zone, shade type, intercropping.

Model	RMSE	MAE	Median AE	Mean MAE per study
Proportional loss, $y = 1 - s$	0.430	0.305	0.210	0.224
One curve for all crops	0.372	0.265	0.193	0.208
Curve per crop type, no pooling	0.313	0.208	0.141	0.165
Curve per crop type, partial pooling	0.312	0.210	0.150	0.162
Partial pooling + covariates	0.311	0.206	0.148	0.161
Gradient-boosted trees	0.321	0.219	0.162	0.184

Table 7: Leave-one-study-out prediction error (RMSE of relative yield, observations with $s > 0$) by crop type: proportional loss, hierarchical curve with partial pooling, and gradient-boosted trees. The last columns give the difference hierarchical minus proportional with a 95% cluster-bootstrap interval over studies (negative: hierarchical better).

Group	Obs./studies	Proportional	Hierarchical	Boosted t
C3 cereals	30/10	0.214	0.187	0.205
Maize	40/10	0.225	0.235	0.248
Grain legumes	58/14	0.215	0.225	0.234
Tubers/root crops	10/2	0.250	0.254	0.257
Four crop types used by the simulator	138/30	0.220	0.222	0.234
Other types (berries, forages, fruits, vegetables)	201/29	0.528	0.361	0.369
All types	339/58	0.430	0.312	0.321

Table 8: Relative yield at 20% and 40% shading: data-driven model (median and 80% bootstrap interval over studies; crop-type curve, covariates at their reference values: mean irradiation, temperate zone, no solar-panel shade, no intercropping) and simulator under the two calibrated ensembles (mean of members; E1 fitted to hand-read curves, E2 fitted to the data-driven curves). The data-driven curve of rice is that of C3 cereals (no rice in the data set). Observations and studies: C3 cereals 41/10, maize 51/10, grain legumes 76/14, tubers 12/2.

Crop	Shading	Data-driven	Simulator, E1	Simulator, E2	Proportional
Wheat	20%	0.93 [0.88, 0.96]	0.88	0.92	0.80
Wheat	40%	0.72 [0.59, 0.78]	0.70	0.77	0.60
Rice	20%	0.93 [0.88, 0.96]	0.85	0.90	0.80
Rice	40%	0.72 [0.59, 0.78]	0.66	0.73	0.60
Maize	20%	0.73 [0.58, 0.86]	0.86	0.90	0.80
Maize	40%	0.42 [0.27, 0.59]	0.69	0.75	0.60
Soybean	20%	0.71 [0.65, 0.79]	0.77	0.74	0.80
Soybean	40%	0.40 [0.34, 0.53]	0.53	0.48	0.60
Potato	20%	0.93 [0.92, 0.97]	0.88	0.92	0.80
Potato	40%	0.71 [0.67, 0.83]	0.73	0.79	0.60

Table 9: Shade-response parameters of ensemble E2, calibrated by gradient descent through the simulator to the data-driven curves and the field trials (12 randomised fits; mean $\pm$ s.d.): radiation-use compensation c and multiplier h on the flowering-window harvest-index penalty (1 = assumed value).

Crop	c	h multiplier
Wheat	0.91 ± 0.31	0.74 ± 0.53
Rice	0.61 ± 0.25	0.83 ± 0.33
Maize	0.14 ± 0.11	0.16 ± 0.13
Soybean	0.05 ± 0.09	1.66 ± 0.38
Potato	0.62 ± 0.20	0.90 ± 0.50

Table 10: Land-equivalent ratio of rules and learned controllers under four true shade responses, with the array density designed under the true response (split 0, baseline climate, floor $\rho = 0.9$, every sixth held-out site, 21 sites). The proportional (conservative) and harsh responses lie outside the range of the calibrated ensemble.

Method	Proportional	Calibrated	Field-fitted	Harsh
AV static	1.052	1.059	1.254	1.037
Seasonal sharing	1.133	1.156	1.385	1.097
Stress rule	1.129	1.160	1.406	1.086
Feedback rule	1.139	1.164	1.392	1.089
Phenology rule	1.164	1.190	1.456	1.122
SALS (trained on proportional response)	1.178	1.209	1.489	1.132
SALS-robust (ensemble-trained)	1.179	1.204	1.491	1.127
SALS-adaptive (ensemble, biomass feedback)	1.176	1.202	1.495	1.124

Table 11: Array design under shade-response uncertainty (ensemble E2), pooled over the three spatial splits (93 held-out sites, every third site). Truth: each of five ensemble members in turn. Designs: oracle (density chosen under the truth), ensemble mean (density chosen under the mean response) and uncertainty-aware (density chosen so that the floor is met in at least 75% of the *other* four members). Means over members; site compliance is the share of sites whose mean retention is ≥ 0.9 .

Design	Method	GCR	Retention	Seasons ≥ 0.9 (%)	Sites ≥ 0.9 (%)	LER
oracle	Phenology rule	0.311	0.924	78	87	1.488
oracle	SALS	0.309	0.930	77	83	1.546
ensemble mean	Phenology rule	0.324	0.919	65	70	1.504
ensemble mean	SALS	0.322	0.925	64	67	1.564
uncertainty-aware	Phenology rule	0.299	0.928	74	78	1.467
uncertainty-aware	SALS	0.295	0.934	73	77	1.521

Table A.12: Paired comparison of SALS with each baseline on the 92 held-out sites of the main split (unit of analysis: site mean over test seasons). Δ LER: mean difference (SALS minus baseline) with cluster-bootstrap 95% CI; p : two-sided Wilcoxon signed-rank test; % better: share of sites where SALS has the higher LER; Δr : difference in mean yield retention with its Wilcoxon p -value.

Climate	Baseline	Δ LER [95% CI]	p	% better	Δr	p	Sites
Baseline	AV static	+0.205 [+0.176, +0.236]	$< 10^{-4}$	100	-0.011	$< 10^{-4}$	92
Baseline	Seasonal sharing	+0.088 [+0.068, +0.109]	$< 10^{-4}$	97	+0.001	0.922	92
Baseline	Phenology rule	+0.038 [+0.025, +0.053]	$< 10^{-4}$	84	+0.001	0.723	92
Baseline	Stress rule	+0.096 [+0.071, +0.124]	$< 10^{-4}$	93	-0.002	0.055	92
+1.5 °C	AV static	+0.208 [+0.180, +0.237]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+1.5 °C	Seasonal sharing	+0.095 [+0.077, +0.114]	$< 10^{-4}$	98	+0.001	0.812	92
+1.5 °C	Phenology rule	+0.033 [+0.022, +0.045]	$< 10^{-4}$	86	+0.002	0.855	92
+1.5 °C	Stress rule	+0.108 [+0.083, +0.134]	$< 10^{-4}$	98	-0.001	0.041	92
+2 °C	AV static	+0.209 [+0.181, +0.238]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+2 °C	Seasonal sharing	+0.097 [+0.079, +0.115]	$< 10^{-4}$	97	+0.001	0.876	92
+2 °C	Phenology rule	+0.032 [+0.022, +0.044]	$< 10^{-4}$	85	+0.002	0.496	92
+2 °C	Stress rule	+0.115 [+0.089, +0.142]	$< 10^{-4}$	98	-0.003	0.016	92
+3 °C	AV static	+0.219 [+0.190, +0.250]	$< 10^{-4}$	100	-0.009	$< 10^{-4}$	92
+3 °C	Seasonal sharing	+0.108 [+0.089, +0.129]	$< 10^{-4}$	96	+0.001	0.935	92
+3 °C	Phenology rule	+0.034 [+0.022, +0.046]	$< 10^{-4}$	86	+0.002	0.330	92
+3 °C	Stress rule	+0.131 [+0.102, +0.162]	$< 10^{-4}$	98	-0.002	0.142	92